

Review - Week 15 Friday

STAT 109: Introductory Biostatistics

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Project portfolio review day (Projects 1-5)

Today we review the full sequence of course projects and connect methods, interpretation, and communication.

Session objectives

By the end of class, you should be able to:

1. Summarize the research question and design for each project.
 2. Identify what each project can/cannot claim (generalization and causation).
 3. Explain the core method used in each project and why it fit the data type.
 4. Interpret key outputs (estimates, p-values, intervals, model fit).
 5. Reflect on one major strength and one limitation for each project.
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Project-by-project review template

Use this same checklist for **Projects 1, 2, 3, 4, and 5**:

1. **Question:** What was the main statistical question?
 2. **Design:** SRS/convenience? RCT/observational?
 3. **Variables:** Response and explanatory variable types.
 4. **Method:** Plot/test/model used.
 5. **Result:** Main numerical/graphical finding.
 6. **Inference:** What claim is justified?
 7. **Limitations:** Bias, confounding, missingness, assumptions.
 8. **Communication:** One sentence rewrite in clear, ethical language.
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Suggested in-class structure

Round 1 (small groups)

- Spend 8-10 minutes per project using the template.
- Assign one speaker and one recorder per group.

Round 2 (whole class share-out)

- Each group presents one project highlight:
 - best methodological choice,
 - biggest interpretation pitfall,
 - one improved conclusion sentence.

Round 3 (individual reflection)

- Write a brief final reflection:
 - Which project best demonstrates your statistical growth?
 - Which concept still needs review before final assessments?
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Common cross-project pitfalls to watch for

- Overgeneralizing from convenience samples.
 - Claiming causation from observational associations.
 - Ignoring missingness/dropout and potential bias.
 - Reporting p-values without checking assumptions.
 - Confusing statistical significance with practical importance.
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End-of-class deliverable

Submit a one-page project synthesis:

- 5 bullets (one per project) with question, method, and conclusion.
- 1 paragraph on how your inference language improved across the term.
- 1 action item for strengthening your final Project 5 report.